

CHAPTER THIRTEEN

Special Considerations for Mountain Running

When you plan to undertake a mountain running event, you will probably know its distance and its elevation gain and loss. Knowing this allows you to use a rather formulaic approach to planning a training progression. Understanding that specificity in your training is important, you should also choose training terrain that as much as possible models the terrain of your target event.

Volume

Below are general guidelines that may help you set realistic goals. Just because you are able to get through one week for each of these example distances doesn't mean that you will be able to log *consistent* weeks at these levels. To build up to even the suggested 50 kilometers (the minimum covered here), weekly distance numbers can take months of continuous running training progression.

Recall from the sidebar “How Kilian Records His Training” on page 127 that he records time, distance, vertical, and the technical difficulty of the terrain. You would be wise to copy this system so that you can learn how the combination of these different components affects you.

Recommended Minimum Peak Weekly Distances for Mountain Races

50K race: Before starting a structured training plan, you should be comfortable with regular weeks of 40 kilometers (25 miles). Peak week = 80–88 kilometers (50–55 miles). Average per week for biggest 12 weeks = 50 kilometers (36 miles).

100K race: Before starting, you should be comfortable with the 50-kilometer training loads (above). Peak week = 120–130 kilometers (74–80 miles). Average per week for biggest 16 weeks = 75 kilometers (48 miles).

100-mile race: Before starting, you should be comfortable with the 100-kilometer training loads (above). Peak week = 145–160 kilometers (90–100 miles). Average per week for biggest 18 weeks = 100 kilometers (75 miles).

Of course, being able to sustain (and tolerate) these peak weekly volumes of training or even exceed them will likely give you better race results.

Seven Recommendations for Trail Racing and Training – Kílian Jornet

I race a lot: I have been doing between thirty and fifty races a year for the last ten years—450 races over the past fifteen years. I don't really prioritize one race over another. I want to do well in all races. But one thing I do is use races to train. It's not that I take it easy during the race; it's that I treat it as intensity work. When I was younger, I did a lot of interval training and strength training, but now I do no strength training and almost no interval training. Instead, I get my speedwork done during races, and when I'm not racing, I do volume. I enjoy volume; it works well for me, and I have a fast recovery so I can assimilate it all.

This is not to say everyone should take this approach. I have been an endurance athlete since I was very young, I worked with a coach for many years, and I studied sports science at university. I have the background to train and race this way.

When people start running (or skimo racing) when they are older, they often train with no structure. There is no long, scientific learning process behind it. And even people who have been running for years can make mistakes in training.

Whether you are new to the sport or are very experienced, take these seven recommendations to heart.

1) Consider the stress of everyday life.

People who come late to endurance sports want to start at a good level, but they often fail to take into account the stressors of everyday life. They train a lot on top of work and family obligations, not realizing that when you train something, it's a stress to your body. And this stress needs recovery time.

2) Don't overdo it. Many people work full-time and train on the side because they really love running. Sometimes these runners get better, get a sponsor, and decide to work less and train much more. Then their performance drops and they burn out because they have become overtrained. Just because you have more time to train more doesn't mean you should fill it all with training.

3) Be realistic about your goals and expectations. Setting a goal—whether it's winning a race, finishing in a certain time,

or just finishing—is what will motivate you to train. It will motivate you to keep pushing through the hard moments and to improve from them. But be humble about what you can do. Honestly analyze yourself in the beginning—your strong points and weak points, even your work situation, diet, and travel. Do not overestimate yourself. To see if something is possible, look at what you want to do and what you will need to change for that to happen.

If a goal is unrealistic, you probably won't reach it, and that can leave you less motivated. What you see as a bad result may actually be a very good result, it just wasn't your *goal* result. In this case, the problem wasn't your performance, it was that the goal was too high.

4) Be progressive. It is important to understand the progression of training and the progression of distances. These days, people want to do a 100-miler, like it's not enough to do a marathon. Maybe they do one 100-miler and it's okay, they get through it. So they do a second one and maybe get through, and then they do a third one and end up broken for five years. You need to move through 10K, 20K, 30K, and beyond. I didn't start out running 100-milers. I progressed over years of structured training.

5) Prepare for the technicality of a course. It is important to think about variables beyond just distance and elevation. People know and understand those, but

they often fail to estimate the technicality of a course—if the tread is soft, grassy, hard, or rocky. We organize a Skyrace near Tromsø, Norway, that has a ridge with some third-class steps, and the downhill is off-trail through a snowfield and rocky terrain. It is not mountaineering, but it is not exactly running. Even if you can run 100 miles with a lot of elevation change, you may not be able to finish a race that is so technically difficult.

6) Train each element, because they are all connected. When it comes to your performance in a race, everything is linked: your physical preparation is linked to your technical preparation, which is linked to your mental preparation and your gear. Each one affects the rest, so they all need to evolve together.

7) Don't take it all too seriously. To keep up motivation, don't take training too seriously. Running and ski mountaineering—these are things we are doing mostly for pleasure and because we like them. It's okay to do things in a serious way, but don't take it too seriously. Train well, but don't make it your whole world. It shouldn't feel like an obligation. That's when it starts to be a problem. I train because I enjoy it.

ATHLETE STORIES

Traversing the Borders of Pain

Emma Roca

Only two weeks had passed since I'd won the 2014 Leadville Trail 100 in Colorado. Now I was set to embark on the Transalpine Run, a stage race done in teams of two that traverses the Alps of Germany, Austria, and Italy. I would be racing in the mixed category with Gerard 'Blacky' Morales, a fellow member of Team Buff. Together we would tackle eight grueling stages for a total of 293 kilometers and 13,700 meters of gain. It would be an unforgettable experience.

Each year the Transalpine changes direction, with the route in 2014 starting in Ruhpolding, Germany, and ending in Sexten, Italy. The course was as challenging as it was beautiful, taking us over mountain passes as high as 3,000 meters and through lush, dreamy alpine valleys—a wonderful landscape. We ran and walked among snowy mountains, navigated technical trails full of rocks and slippery roots, and traced ridges littered with large blocks of granite. We passed under the Tre Cime di Lavaredo and cut through forests full of mushrooms. We crossed borders that went beyond physical and mental limits. We ran day after day with minimal recovery between each stage.

I had unique insight into the impact this race was having on my body. As runners, we all know that these big efforts stress our systems, but as a doctoral student of biochemistry, I have a more scientific understanding of the biochemical and physiological effects of stage races. During this race, in fact, I was conducting a study for my thesis.

Blacky and I were one of eight mixed Spanish teams who offered up our bodies, blood, and urine in the interest of developing a fuller picture of what happens during a multiday effort like the Transalpine. From before the start to after the finish, we underwent numerous tests to determine to what degree we were recovering—or not. It was clear from the results

that our bodies were breaking down progressively from stage to stage.

You would think that having an intimate knowledge of ultrarunning physiology would somehow translate into a more carefully crafted running practice. Or that after fifteen years of racing ultras internationally and winning world championships, I would know what is best for my body. Sometimes that is not the case. As they say, *"En casa del herrero, cuchillo de palo."* (In the blacksmith's house, a wooden knife.) During the Transalpine, the crazy competitor in me overruled the logical scientist.

I hadn't let myself recover enough from Leadville, and the pressure of running as part of a team—a team that was gunning to make the podium—placed too many demands on my already overtaxed body. It was especially challenging being a woman in a race with a man. The physical and psychological tiredness flowed easily, and I went through the occasional moment of tension with Blacky. That said, he and I worked very well together overall, with him helping me a lot through each stage.

At the end of the first day, I noticed a pain in my left knee that increased as the days went by. Every morning I looked for a hotel near the staging area that had a stationary bike in its gym so I could warm up. Even then, I started each stage at a half-limp, hobbling until the knee warmed up and I could get into a rhythm that allowed me to finish the stage and even maintain our lead. But the pain at the end of each day was intense: the tears came and I could barely walk. After the competition, an MRI confirmed that I had a stress fracture in the head of my femur—an injury that would keep me immobilized for six months.

We won the Transalpine that year, despite my stress fracture. Perseverance, tenacity, commitment,



Emma Rocca cools off in celebration of her team's victory at the end of the final stage of the Gore-Tex Transalpine Run. Sexten, Italy. Photo: Harald Wisthalei

and competitiveness are great stimulants for me, and in this race they pushed me past the pain and into first place. It took strength and determination to finish and win, even at the risk of exacerbating my injury.

When you live, analyze, and feel all the wear, pain, and impact of the kilometers of a race like the Transalpine—not only physically but also mentally—you ask yourself the obvious question: Why do we keep going when our bodies tell us to stop? I do not have a rational answer to offer. It is true that endorphins exist, and sometimes the suffering in the moment becomes its own reward later on, once you've forgotten just how bad it was.

I know what is happening inside of me during a stage race, and I know how much my body needs to rest, but the experiences I live in an ultra or a long mountain race are priceless mental rewards. I'd rather pay the price that comes with running ultras than miss out on these unique, extraordinary adventures. ■

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Structuring a Category 1 Athlete's Mountain Running Training Plan

If you fit into this group, the biggest single thing you can do to improve your race performance is to train *more*. Not train harder, just train more hours, accumulate more vertical meters, run more kilometers. That's because your performance in the sports covered in this book is almost entirely dependent upon your basic aerobic capacity to do work. By now you understand that the only way to improve that quality is to do more low-to-moderate-intensity training (Zones 1–2). Elevating this basic aerobic work capacity is the key to future success. So, that's where the emphasis has to be placed. There is no shortcut, no secret workouts. You need to first condition your body to handle a lot of training; then later as you become fitter, you can start to layer on higher-intensity work that will allow you to run or ski faster for longer. But right now if you can't run continuously for an hour or two, on mountainous terrain, day in and day out, you do not need to worry about high intensity. This is a bottom-up approach where each subsequent layer of training at higher intensities stands upon the base provided by the levels below and supports the levels above.

For athletes in this category, the need for a very gradual ramping up of weekly volume must be strongly emphasized. The progression of training volume that we suggest (see fig. 11.7 page 291) is on the conservative side for a reason. By far, the most common running injuries are to the lower legs and are the result of increasing running volume too fast. These are typically overuse injuries caused by an accumulation of the chronic stress of impact. The problem with them is that by the time you feel the injury, there is likely to have been a lot of damage done and you may lose weeks to recovery. Avoiding injuries of this type has to be a first priority of any training plan. Once you are injured, no plan will work for you.

The connective tissue in your legs will be slow to adapt to the stress of running; the best advice is to proceed with caution when increasing running volume and intensity. It takes weeks of sustained and repeated running for the tendons and fascia to increase in elasticity and strength to handle the pounding of thousands of impacts each day, each of which will be at least two times your body weight and much more on the downhill. Patience is a virtue when it comes to increasing running volume. Be kind to your body and allow it time to adapt before demanding too much from it.

A 50K race is a great way to dip your toe into ultras. With that in mind, the first part of this section will address the specific needs of the less experienced runner, presumably with less running history and fewer miles in the legs. For them, a twenty-week plan is what we consider as the minimum training period if preparing for a first-time 50K race.

If you have little recent running history, you'd be wise to precede this program by at least a six-week Transition Period (see page 145). However, if you have been



Urban Zemmer putting a big base to the test on the final section of the 2015 Crepa Neigra Vertical-K race in Canazei, Italy. Photo: Martina Valmassoi

running regularly about 20 to 25 miles (35 to 50 kilometers) per week, you can start with week 1. Don't expect optimal results if you spent the winter on the couch and then try to jump right into a 40-mile week. Even if you have been skiing all winter, you'll need at least a short Transition Period to get your running legs back. Otherwise, you may end up injured in a few weeks.

Keep the following critical aspects in mind when designing a training plan. These will help you create good training plans and are discussed in more detail in the Methodology Section.

- You will not really begin running training until you have accumulated about 100 miles (62 kilometers) in your legs. Until then you are just conditioning your legs so that they can handle daily running. It might take some seasoned runners two weeks to hit this target early in their training cycle; it might take beginners at least six weeks.
- Intersperse big training weeks with lower-load weeks so that your body has a chance to absorb and adapt to all the good training of the build period. Do not hold the training load constant for multiple weeks. A lack of modulation will cause stagnation and possibly overtraining.







Running with sheep in Italy is a lot safer than running with bulls in Spain. Mortirolo Pass, Camonica Valley, Italy. Photo: Davide Ferrari

- The principle of specificity as applied here means that you need to run (and hike some steep sections) for training. Biking and swimming are great exercises, but they should not be substitutes for running unless injury necessitates. These can be effective recovery workouts to give tired legs a break. You should also choose terrain as similar to your goal event as possible. If you are going to race on rocky or rough trails, that will demand very different coordination than an event that takes place on smooth trails. Incorporate frequent short sections of fast downhill running to develop the strength and quickness needed, especially on very rough trails.

Example Weeks for Category 1 Mountain Runners

Use the following example weeks to help you understand the general layout of a week's structure.

Previous page: Jeff Browning running past Island Lake, Colorado, during a recon of the Hardrock 100 course. Jeff won this race in 2018. Photo: Steven Gnam

	AM	PM	Comments
Day 1	Rest day.	Off.	This can be a good day to take off or use for light recovery activities after a big weekend of training.
Day 2	Z1: up to 15% of weekly vertical. Z2 if you suffer from ADS.	Core and general strength.	Choose the appropriate strength stage from your assessment.
Day 3	Z2: up to 15% of weekly vertical.	Off.	
Day 4	Z1 or recovery: at least 15% of weekly time.	Optional ME. See page 240 in the Strength section.	Stay in the zone. Done on flat-to-gentle terrain.
Day 5	Hill Sprints, 6–8x10 sec.	Off.	When appropriate, move to Hill Sprints for strength.
Day 6	Z2: up to 15% of weekly vertical.	Optional second workout: Z1 or recovery. Up to 5% of weekly vertical.	
Day 7	Z1: up to 40% of weekly vertical.	Off.	Z2 if you suffer from ADS.

Fig. 13.1 Example of an Early Base Period Week for a Category 1 Mountain Runner

Note: In these early weeks (at least eight weeks) the main goal is to accumulate running time. Keeping the intensity low will allow you to get more distance in. This conditioning period is critical for a successful and injury-free implementation of the rest of the plan. Start work on ME workout for general strength if you are deficient.

	AM	PM	Comments
Day 1	Rest day.	Off.	This can be a good day to take off or use for light recovery activities after a big weekend of training.
Day 2	Z1 or recovery: up to 10% of weekly time.	Core strength.	Stay in the zone. Done on flat-to-gentle terrain. If still feeling the weekend in your legs, then make this a very easy recovery run.
Day 3	Z1-Z3: up to 15% of weekly vertical. Up to 5-8% in Z3.	Off.	Start with 3x8 min. Increase Z3 volume only as your body allows.
Day 4	Z1: up to 15% of weekly vertical.	Off.	Z2 if you suffer from ADS.
Day 5	Hill Sprints, 6-8x10 sec.	Optional second workout: Z1 or recovery. Up to 5% of weekly vertical.	
Day 6	Z1: up to 15% of weekly vertical.	Off.	Z2 if you suffer from ADS.
Day 7	Z1: up to 40% of weekly vertical.	Off.	Z2 if you suffer from ADS.

Fig. 13.2 Example of Late Base Period Week for Category 1 Mountain Runner

Note: These weeks see the introduction of high-intensity training along with increasing volume. Reduce or eliminate the ME training if and when you start the Zone 3 work.

	AM	PM	Comments
Day 1	Rest day.	Off.	This can be a good day to take off or use for light recovery activities after a big weekend of training.
Day 2	Z1 or recovery: up to 10% of weekly time.	Core strength.	Stay in the zone. Done on flat-to-gentle terrain. If still feeling the weekend in your legs, then make this a very easy recovery run.
Day 3	Z1-Z3: up to 10% of weekly vertical. Up to 5% in Z3.	Off.	Start with 3x8 min. Increase Z3 volume only as your body allows.
Day 4	Z1: up to 10% of weekly distance and vertical.	Off.	Z2 if you suffer from ADS.
Day 5	Hill Sprints, 6-8x10 sec.	Off.	
Day 6	Z1: up to 30-40% of weekly distance and vertical.	Off.	Z2 if you suffer from ADS. In this week you are doing back-to-back big days to more specifically prepare for longer runs.
Day 7	Z1: up to 30-40% of weekly distance and vertical.	Off.	Z2 if you suffer from ADS. In this week you are doing back-to-back big days to more specifically prepare for longer runs.

Fig. 13.3 Specific Week for the Late Base Period

Note: Use of back-to-back long runs to simulate more of the demands of the actual event.

Structuring a Category 2 Athlete's Mountain Running Training Plan

If you meet that second set of criteria listed on page 287—have done many successful races and been logging a lot of distance and vertical for two to three seasons—you no doubt have established a solid base of training. You may no longer be satisfied with just finishing these races. Or you may be interested in moving up to a longer race, or are looking for ways to boost your performance. You may even be planning on an event that includes much more elevation gain and loss. If so, refamiliarize yourself with the following guidelines.

Moving Up in Distance

If you are planning on moving up to a significantly longer race than you have done, either in the past six months or ever, such as moving from a 100K race to a 100-mile distance, then the way forward is simple: You must increase the volume of running. Instead of maintaining typical weeks of 40 to 50 miles (60 to 80 kilometers) with peak weeks of 60 to 75 miles (100 to 120 kilometers), you will need to gradually build your typical weekly volume into the 70 to 80 miles (112 to 128 kilometers) range with peak weeks of 100 to 110 miles (160 to 180 kilometers).

For racers finding themselves in this position, try to follow a similar volume progression as presented in the example weeks fig. 11.7 on page 291. Similar to when you were building up to your first long-distance race years ago, concentrate your efforts first on accumulating the volume of training so you feel confident you can cover the distance and then add intensity sparingly and conservatively.

Refer to the charts on page 188–293 for idealized volume progression plans for three different scenarios:

1. A single A priority 100K race with a B priority 50K race earlier.
2. A single A priority 100-mile race with a B priority 100K race earlier.
3. Several C and B priority races leading up to an A priority 100K race.

Racing More Often or Improving Performance

If you are not planning to move up to longer races than your current training base will support, but instead are looking to improve performance or race more often throughout the season, consider the addition of more high-intensity training to your program. This example week will be appropriate for those runners with a solid aerobic base who do not have ADS and can pass the Ten Percent Test (see page 91).

	AM	PM	Comments
Day 1	Rest day.	Off.	This can be a good day to take off or use for light recovery activities after a big weekend of training.
Day 2	Z1 or recovery: up to 10% of weekly time.	Core strength.	Stay in the zone. Done on flat-to-gentle terrain. If still feeling the weekend in your legs, then make this a very easy recovery run.
Day 3	ME. See page 240 in the Strength section.	Off.	Use this in place of Z3-Z4 intervals for much of the Base Period.
Day 4	Z1: up to 10% of weekly distance and vertical.	Off.	Z2 if you suffer from ADS.
Day 5	Z1: up to 10% of weekly distance and vertical.	Off.	
Day 6	Z1: up to 30% of weekly distance and vertical.	Off.	Z2 if you suffer from ADS. In this week you are doing back-to-back big days to more specifically prepare for longer runs.
Day 7	Z1: up to 30-40% of weekly distance and vertical.	Off.	Z2 if you suffer from ADS. In this week you are doing back-to-back big days to more specifically prepare for longer runs.

Fig. 13.4 Example of a Category 2 Mountain Runner Specific Base Period Week

Note: This athlete wants to boost volume and simulate the demands of the longer event week.



Megan Kimmel tries to recover after winning the 2017 Marathon du Mont Blanc, Chamonix, France. Photo: Martina Valmassoi

	AM	PM	Comments
Day 1	Rest day.	Off.	This can be a good day to take off or use for light recovery activities after a big weekend of training.
Day 2	Z1-Z3: up to 15% of weekly distance and vertical. Gradually add up to 5% in Z3.	Core Strength.	Use intervals, tempo runs of progressive distance here. Increase Z3 volume only as your body allows.
Day 3	Z1 or recovery: up to 10% of weekly time.	Off.	Stay in the zone. Done on flat-to-gentle terrain.
Day 4	Hill Sprints, 6-8x10 sec along with Z1. Up to 10% of weekly distance.	Off.	
Day 5	Z1-Z4: up to 10% of weekly distance and vertical. After 4-6 weeks of Z3, gradually add up to 5% in Z4.	Off.	Once comfortable with the volume of Z3 (up to 10% of weekly), add Z4 intervals starting with 30/30s.
Day 6	Z1: up to 15% of weekly distance and vertical.	Off.	
Day 7	Z1: up to 40% of weekly distance and vertical.	Off.	

Fig. 13.5 Example of a Category 2 Mountain Runner Late Base Period Week

Note: This athlete wants to run more and shorter races in one season.

NOTES ON ADDING MORE INTENSITY

- The type of intensity you add to your program depends on the event you are targeting. Long Zone 3 sessions will serve well for long-distance races. On the other hand, shorter Zone 4 and Zone 5 sessions will be very useful when you are training for a VK or shorter race.
- In general, use a mix of intensities in the overall plan. Opinions vary widely, but there is no conclusive evidence to support one method over others when it comes to building endurance. Fitter athletes will benefit from longer duration repetitions (more than six minutes) and higher overall volume of high intensity.
- The volume of high-intensity training in each week, and within each workout, should not start too high and should increase gradually. Add only as much as you can before it forces you to drop the volume of your aerobic base (Zones 1 and 2) training by more than 5 percent. If you can't maintain your volume of Zones 1 and 2 training, you risk having a short peak followed by a decline in performance.
- Drop all Zone 2 training when you begin adding Zone 3 work to your plan. Do all aerobic base work in Zone 1. Zone 2 will be too stressful for most athletes when they begin adding intensity.
- Include one Hill Sprint workout (see page 182) in most weeks to build and maintain specific leg power.
- If you need to improve your muscular endurance (see page 240), begin to add the ME workouts in the early weeks and drop them when you introduce Zones 3 and 4 workouts. Add these workouts after at least four weeks of only Zones 1 and 2 base training. Use them one time per week for eight weeks. The Zones 3 and 4 intensity later will maintain the muscular endurance you build.
- After six to eight weeks of successfully incorporating these Zone 3 workouts, it can be time to introduce some Zone 4 work to the week. Replace some of the Zone 3 with half the volume of Zone 4. Thirty minutes of Zone 3 would be replaced by fifteen minutes of Zone 4. As mentioned above, drop any Zone 2 work when adding Zones 3 or 4 work. All aerobic base maintenance will be done in Zone 1 or even lower at a Recovery effort. Kilian's favorite way to introduce Zone 4 is with the use of 30/30s (see page 177).

ATHLETE STORIES

Wildlife Rendezvous

Jeff Browning

During Colorado's wild and tough Hardrock 100-Mile Endurance Run, I descended the narrow, rock-strewn Continental Divide Trail as it parallels cascading Pole Creek. It was the middle of a clear and beautiful night, and the moon was waxing gibbous. I was on pace to break the double record—the combined times of the Western States 100 and Hardrock 100, two classic races held a mere nineteen days apart. As I was taking in the gorgeous silhouettes of the surrounding rock formations and the rush of the creek just to my left, I caught glowing eyes in my headlamp to my right.

As with all nighttime mountain running excursions, it's a solid strategy to stop and identify what genus those eyes belong to. I flipped the headlamp to its highest setting—a hundred-meter flood that illuminated not one but two mountain lions lounging on a slab of rock above me. A quick assessment of the situation revealed maybe 30 meters of brush between the trail and where the slab of rock jutted up.

The one in front, the bigger of the two, stood up. I yelled in a deep, commanding voice, "You don't want any part of me!" I then proceeded to keep my light on them and walk down the trail, beating my chest and yelling like a deranged wild man. The big one paralleled me on the rock. Lucky for me, her slab quickly ran out, and the trail curved around a wall of rock that shielded me from their line of vision.

As soon as I was out of sight, I turned and bombed the downhill like I was racing a 10K, all fatigue from 80-plus mountainous miles completely gone. Every 50 to 100 meters, I stopped, turned, and scanned behind me for eyes—nothing. Then I took off again like I'd stolen something. I continued this pattern, with less frequent looking behind me, for the next twenty minutes.

Trail running in wild, mountainous locales is fairly stealthy, especially if you're running solo. It has set me up for some very unique appointments with wildlife, and cats have been a particularly good motivator to get my mind out of the pain cave.

The Stalker and the Stalked

That Hardrock encounter was my third rendezvous with a mountain lion in the night portion of a 100-mile race. The second was uneventful, but the first and most harrowing occurred during the San Diego 100 at mile 97. It had already been a creature-filled night: my pacer, Jesse, and I had marveled at a tarantula on the Pacific Crest Trail at mile 83—our own personal live biology session—and we later followed a stubborn skunk for a quarter mile before he hissed and took off into the tall grass.

Just a mile after Jesse and I sprinted by that skunk to avoid getting sprayed, I caught movement to my left. Something had jumped toward us off a car-size boulder and into the tall grass. I stopped and popped my headlamp to high beam. A cat raised its head out of the grass, maybe 20 meters away. Two green eyes glowed in the night.

"Cougar, cougar!" I yelled to Jesse, to which he frantically replied, "Where?" and quickly, "I see it!"

We began yelling and kept the cougar blinded with our lights. It was within the dreaded pouncing distance; one spring from those powerful hind legs and we'd be dinner. We made our way sideways, walking down the trail with our lights fixed firmly on the cougar. The cat ducked down and disappeared into the tall grass.

"Where is it?" Jesse yelled.

It emerged adjacent to us, paralleling our movements then crouching under a half-fallen tree. The cat's ominous green eyes watched our every move.



Jeff Browning just after finishing the Run Rabbit Run 100 in the predawn hours. Steamboat Springs, Colorado.
Photo: Fredrik Marmsater

After we'd gained enough distance and finally started to run again, we continued to pause every 40 meters to scan behind us and to our sides to be certain the cat wasn't tracking us. As frightening as it was in the moment, now it's another special story to tell.

Hurdling and Trail Running

Not all of my wildlife encounters have been dangerous. During the Bighorn 100 in northern Wyoming, as I was bombing down to Little Bighorn Canyon, where the trail plunges more than 2,000 feet in 2 miles, the scrub oaks erupted to my right. I'd been fully engaged in the enterprise of negotiating technical terrain, and now I panicked. The sound had my imagination racing through all the possibilities: bear, moose—I'm gonna die!

Instead, a marmot came scurrying out and intersected my path at top speed. I screamed like a three-year-old who just lost a cookie and hurdled the critter without breaking pace. With my heart rate spiked, and relieved that it wasn't a bear, I shouted over my shoulder as I descended: "You scared the crap out of me!"

Head Protection Needed

One of my favorite wildlife encounters is also one of my most bizarre experiences in ultrarunning. I was pleased to be leading the Cascade Crest 100 in central Washington at mile 52. Running up Keechelus Ridge at dusk, I was buzzed by a barred owl about three feet over my head. The owl swooped up into a fir tree and stared down at me with those big, wise eyes. A line immediately came to mind from the children's book *Sam and*

the Firefly by P.D. Eastman, which I regularly read to my young kids at the time. I quoted Sam the owl's nightly call to action: "Who? Who? Who wants to play?" Chuckling at my wit, I proceeded to run up the road.

As soon as my back was turned, the owl dive-bombed me and clawed the back of my head—*whap!* My chin flew to my chest, and I instinctively checked my head for blood. I guess he did want to play! I aimed a few handfuls of gravel around where he was perched, and he flew off. I continued up the road, but constantly looked over my shoulder for signs of Sam.

Running in big mountain landscapes and backcountry wild places is meditative and special, but understanding the dangers is also important—especially when it comes to wildlife. It's important to research the types of wildlife you might encounter where you will run and the strategies for dealing with those creatures. Run like hell, back away slowly, or hurdle the marmot. Whatever the encounter, knowing how to act is the first step in walking away safely with a good story to tell later. ■

Jeff Browning, aka 'Bronco Billy,' is a professional ultrarunner and online endurance coach. He loves the flow and solace that running in big, wild landscapes offers. Embracing the space (but never the carnivorous wildlife) is what has kept him running and competing in ultramarathons for nearly two decades. You can find out more about his coaching services or running escapades at GoBroncoBilly.com or follow him on Twitter and Instagram @GoBroncoBilly.

Moving Up in Vertical

The amount of vertical feet (or meters) you include in your weekly training is going to depend on several factors: the vertical in your goal event, your access to hills and mountains for training, and your training history. For the best results, select your goals according to what you have available to train on, and then train on hills similar to the goal race.

If you've been counting on the vertical meters in your weekly training to build the specific strength you need, but are feeling that your legs are the weak link, consider adding a period of ME training to your program.

NOTES FOR WEEKLY VERTICAL

- Begin your vertical progression with no more than a half of your goal event's total vertical in your first week.
- Incorporate muscular endurance work into the plan from the start. Keep it up as long as you are seeing gains. See page 240 in the Strength Section for more details.
- Plan on training a minimum of your goal event's total vertical during at least two of your precompetition period weeks.
- These weeks should correspond to the bigger distance weeks.
- Do not increase distance, vertical, and intensity in a single week. Try to increase only one at a time. If you need to increase two of these, then keep the increases to under 15 percent. Example: 10 percent distance increase with 5 percent vertical increase, or 8 percent increase in Zone 3 time with 7 percent increase in distance.
- As with distance, increase vertical no more than 10 percent in consecutive weeks and no more than 20 percent in a four-week span (three building weeks followed by a recovery week).
- Cut vertical by a similar amount as distance reduction during recovery weeks (40–60 percent).

	AM	PM	Comments
Day 1	Rest day.	Off.	This can be a good day to take off or use for light recovery activities after a big weekend of training.
Day 2	ME and core strength. See page 240 in the Strength section.	Recovery activity.	
Day 3	Z1 or recovery. Stay in the zone. Up to 10% of weekly vertical.	Off.	Best if foot borne, outdoors on hilly terrain. But laps in a tall building or time on a treadmill or stair machine can be substituted.
Day 4	Z1 or recovery. Stay in the zone. Up to 10% of weekly distance and vertical.	Off.	Best if foot borne, outdoors on hilly terrain. But laps in a tall building or time on a treadmill or stair machine can be substituted.
Day 5	Z2 vertical: up to 15% of weekly vertical.	Off.	Best if foot borne, outdoors on hilly terrain. But laps in a tall building or time on a treadmill or stair machine can be substituted.
Day 6	Z1: up to 25% of weekly vertical. Can be Z2 if suffering from ADS.	Hill Sprints, 6-8x10 sec.	Best if foot borne, outdoors on hilly terrain. But laps in a tall building or time on a treadmill or stair machine can be substituted.
Day 7	Z1: up to 40% of weekly vertical. Can be Z2 if suffering from ADS.	Off	Best if foot borne, outdoors on hilly terrain. But laps in a tall building or time on a treadmill or stair machine can be substituted.

Fig. 13.6 Example of a Category 2 Mountain Runner Early Base Period Week

Note: By incorporating a ME workout, this athlete is aiming to complete a goal event with much more vertical than has been typical for this runner.



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ATHLETE STORIES

The Happiness Factor

Emelie Forsberg

I do not run to compete. I do not train to win races. I run because it brings me joy.

I came into the world of racing later than most. I was very active as a child, doing all kinds of sports, but I stopped everything when rock climbing stole my heart when I was fifteen years old. After that, I devoted all my free time to climbing—first trad and later sport. Running and skiing were still there for me as old friends, providing my endurance training, but I identified as a climber.

It was not until I was twenty-two that I realized that running had become my new passion. I was a devoted runner; I loved it so much. Now it wasn't climbing I built my life around, it was running. On foot, I explored the mountains surrounding where I lived, worked, and studied.

The simplicity of running appealed to me: I loved that I could just grab my shoes and go out to get some fresh air, clear my mind, and be in nature. I circled mountain massifs to see what was on the other side, and I headed up unfamiliar trails, curious where they would take me. I ran up to and around summits. I ran to see how many kilometers my legs could cover.

I did maybe two races every year, and I liked the racing because I met so many people who loved the same thing I loved. And I enjoyed pushing myself a little bit harder than I would have done if I was alone. I never thought that one day I would be a professional mountain athlete. I was just doing what gave me so much happiness: running (plus skiing in the winter when I couldn't run in the snowy mountains).

When I was twenty-five, I slipped on a banana peel into the world of full-time racing. I had just joined the Swedish Salomon team (Free shoes! Race fees!), and the international team invited me to a training week,

which led to my trying my first international race: the 2012 Dolomites Skyrace.

I was amazed at how many runners showed up for the event. I was a little nervous at the start and tried to follow two strong ladies up the first climb. I was third to the summit, then first across the finish line with a new course record. I loved every step of it and wanted to do more.

The decision to become a professional runner was not easy for me. I was almost finished with my biology studies when I got the opportunity to travel the world running races. I chose the running, and it worked. But I was and always will be conscious of why I'm doing this: my top priority is to not lose the happiness—the feeling I had when I chose to build my life to be able to run. It was not for winning a race, it was for loving my life.

For me racing has never been and will never be the number one thing. To win a race is big, but it's not everything. This is only my approach. I have met many athletes whose priority is to train toward the goal of winning "A" races, no matter what. I admire their determination.

We all need to find our own way to keep the motivation to train, because in the end, that is what makes us stronger. For me, racing should be fun. But I won't sacrifice my health and happiness for it. It's a luxury for me, and it's fun to push myself to be the strongest version of myself. ■

Emelie Forsberg is a devoted lover of mountains, a small farmer, a proud vegan, and an avid baker.

